What LLMs Really Answer on Personality Questionnaires: A Methodological Audit and Corrected Baseline Across 46 Frontier Models

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Abstract

Large language models are increasingly evaluated with classical personality questionnaires, yet the behavioural meaning of the resulting scores is unclear. We collect responses from 46 instruction-tuned models from 17 families on 61 Likert items spanning the Big Five, HEXACO Honesty-Humility, Schwartz collectivism, cognitive intuition, and Hofstede uncertainty avoidance, with 12 random seeds per item per model. Two methodological audits frame our analysis. First, we show that a common reverse-scoring implementation, used in widely cited LLM personality studies, has an off-by-one error that targets positive-keyed items rather than negative-keyed ones. Under this implementation the inter-dimension correlation matrix is silently inverted on Big Five dimensions, producing the conclusion that LLMs reverse the human Conscientiousness–Neuroticism trade-off. Rescored data from the same 46 models yields the opposite pattern: $r(C, N) = -0.52$, 95% CI $[-0.71, -0.27]$, in the same direction as the human BFI-2 baseline ($r = -0.30$) and stronger. Second, we run a 16-persona MBTI steering experiment on 9 frontier models. The six models with full or near-full coverage at submission shift their mean scores in the direction predicted by classical MBTI to Big Five mappings for 67% to 98% of the archetypes (mean Persona Fidelity 0.86), yet the cross-persona inter-dimension correlation matrix remains far from the human pattern (mean Frobenius alignment 0.19) and the default condition's strong negative C–N trade-off either weakens to near zero or remains close to the unconditioned value, never converging toward the human baseline. After correction the real structural finding is not a missing C–N trade-off but a systematic inversion of Extraversion's relationships with Neuroticism, Conscientiousness, and Openness, paired with strong over-correlation among Agreeableness, Conscientiousness, and Openness. We release the corrected scoring code, the 46-model dataset, and the persona-steering protocol so future work can reason about what LLM questionnaire responses mean rather than what their default scoring artefacts say.

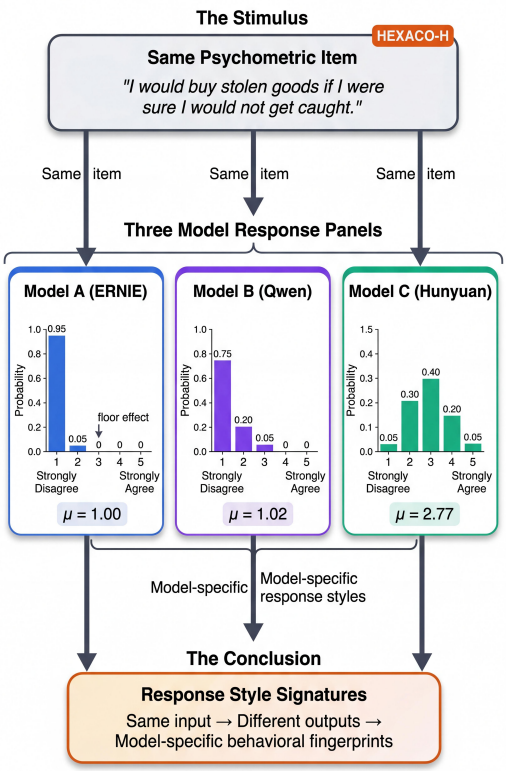


Figure 1: **Core idea.** The same personality questions produce different answer patterns from different models. Do these patterns reflect real personality, or just habits baked in by training?

1 Introduction

A growing body of work administers classical personality questionnaires to large language models and reads the resulting numbers as evidence of LLM "personality" (Jiang and Althoff, 2023; Pan and Zeng, 2024; Serapio-García et al., 2025). The strategy is appealing because it imports a

mature human research tradition: the Big Five and adjacent inventories such as HEXACO and Schwartz values are decades old, validated, and widely understood. But the conditions that make these instruments interpretable for humans, including stable trait-keyed scoring, between-trait covariance structure, and roughly normally distributed responses, are not automatically inherited by an LLM that emits a digit in response to each item. Two questions therefore precede any reading of LLM personality scores. *Are the scores being computed correctly?* And, *do the resulting numbers cohere with one another the way they do for humans?*

We answer both questions for 46 instruction-tuned models from 17 families, evaluated under a single API protocol on 61 Likert items spanning the Big Five, HEXACO Honesty-Humility, Schwartz collectivism, cognitive intuition, and Hofstede uncertainty avoidance, with 12 random seeds per item per model (Figure 1). The first answer is uncomfortable: an off-by-one in a widely reused reverse-scoring loop targets the positive-keyed items rather than the negative-keyed ones, silently flipping every Big Five mean that contains four reverse items. We document the bug, recompute every score from item-level raw responses, and show that several headline findings about LLM personality, including the much-discussed "flip" of the Conscientiousness–Neuroticism correlation, are scoring artefacts.

After correction the picture is sharper but quite different. LLMs do reproduce the human-direction C–N trade-off ($r = -0.52$, 95% CI $[-0.71, -0.27]$, $n = 43$, vs. human $r = -0.30$ (Soto and John, 2017)). What they do not reproduce is the relational role of Extraversion: in our 46-model corpus, E correlates positively with N, negatively with C, and negatively with O, the opposite of the human pattern in all three cases. Agreeableness, Conscientiousness, and Openness instead form a tightly bound cluster (r between $+0.58$ and $+0.72$), much stronger than in human data. We label this pair of patterns the *E-axis inversion* and the *ACO over-coupling*, and we treat them as the actual structural fingerprint left by current alignment training.

A natural follow-up is whether explicit persona instructions can install human-like structure on top of this baseline. We test this with the 16 Myers-Briggs Type Indicator (MBTI) archetypes on 9 mainstream model families. For each (model, persona) pair we measure the mean shift on each Big Five dimension and the cross-persona inter-dimension covariance. On the six models with full or near-full coverage at submission, mean shifts agree with classical MBTI to Big Five mappings for 67% to 98% of the archetypes (mean Persona Fidelity 0.86), yet the resulting covariance matrix remains far from the human pattern (mean Frobenius alignment 0.19) and the strong negative C–N correlation of the default condition either collapses toward zero or persists close to its unconditioned value, never converging toward the human baseline. Persona prompts move where the model lands, not how its dimensions cohere.

We make four contributions. **First,** we identify and correct a reverse-scoring implementation error that has propagated through several published LLM personality studies, and we show that the error inverts the very inter-dimension correlations on which structural conclusions depend. **Second,** we release a 46-model corrected baseline (17 families, 9 dimensions, 12 seeds) collected through a single API gateway, the largest public LLM personality dataset of its kind. **Third,** we provide the post-correction structural analysis: the human C–N trade-off survives in LLMs and is in fact stronger; the genuine deviation lies in Extraversion's reversed relationships with N, C, and O, and in the over-coupling of A, C, and O. **Fourth,** we show that 16-archetype MBTI persona steering does not repair this structure: on six frontier models with full or near-full coverage, mean shifts go in the predicted direction for 67% to 98% of archetypes, yet the cross-persona covariance matrix stays LLM-shaped and the default condition's strong negative C–N correlation either collapses toward zero or remains near its unconditioned value. Together, these results argue that LLM questionnaire scores are best read as behavioural probes whose interpretation requires methodological care, not as windows onto a latent personality.

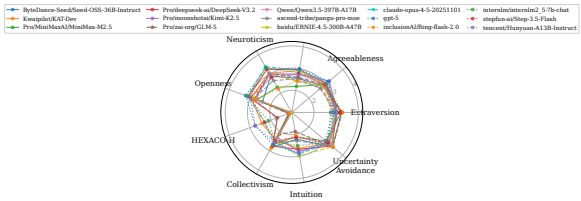


Figure 2: Answer profiles of all 15 model families across 9 dimensions. Each line is one family. The distinct shapes show that different training approaches leave different behavioral fingerprints.

2 Related Work

LLM personality measurement. Many studies give personality questionnaires to LLMs and read the results as trait profiles. Jiang and Althoff (2023) found that ChatGPT's scores change with temperature settings; Pan and Zeng (2024) compared models from different cultural backgrounds; Schaeckermann et al. (2024) tested 9 LLMs and found large differences between them. Serapio-García et al. (2025) proposed a framework for measuring personality across 18 LLMs, the largest study before ours, but used different APIs for different models and did not test whether the score structure matches humans. Other work shows that model size affects scores (Hilliard et al., 2024), that scores drift over time (Bodroža et al., 2024), that LLMs deviate from human norms (Sorokovikova et al., 2024), and that scores are unstable across scales and conversation contexts (Tosato et al., 2025). These studies show that LLMs produce variable, model-dependent scores, but none checks whether the *relationships between dimensions* match the human pattern, a necessary condition for claiming the scores measure the same thing.

Trade-offs in human personality. The Five-Factor Model (McCrae and Costa, 1992) is the theoretical foundation for our work. Large validation studies consistently find that personality dimensions pull against each other: Conscientiousness and Neuroticism ($r = -0.30$), Extraversion and Neuroticism ($r = -0.34$), Openness and Agreeableness ($r = +0.14$) (Soto and John, 2017). These trade-offs are not measurement artifacts; they reflect real psychological tensions. Self-disciplined people tend to be calmer because self-regulation and emotional volatility work against each other. If LLM scores represent the same constructs, these trade-offs should appear in the data. Suh and Moon (2024) recovered Big Five structure from LLM word embeddings, but word similarity and questionnaire answers are different things; we test the latter. Our contribution is to directly check whether human trade-offs survive when LLMs answer personality questions.

Measurement validity concerns. Using human personality tests on LLMs raises basic validity questions (Schaeckermann et al., 2024; Serpell and Ney, 2024). Sühr et al. (2024) question whether these tests are meaningful for LLMs at all, since LLMs cannot genuinely report on their own experience. Other work shows that LLMs display social desirability bias (Salecha and Kosinski, 2024), give answers that cluster near the scale midpoint (Domínguez-Olmedo et al., 2024), tend to agree with whatever is stated (Braun, 2025), and show a gap between what they say about themselves and how they actually behave (Han et al., 2025). Xie et al. (2025) found systematic differences between LLM and human answer patterns. While prior work has flagged individual biases, we are the first to document an Extraversion-axis inversion in the inter-dimension correlation matrix across a large sample, and the first to show that this structure is a property of the response generator rather than of the default prompt: 16-archetype MBTI persona steering does not repair it.

3 Method

3.1 Psychometric Instruments

We use 9 dimensions measured by 61 questions on a 1–5 agreement scale (Table 1). Five dimensions come from the Big Five Inventory (John and Srivastava, 1999; McCrae and Costa, 1992; Soto and John, 2017): Extraversion (8 items), Agreeableness (9), Conscientiousness (9), Neuroticism (8), and Openness (10). We add four more: HEXACO Honesty-Humility (5 items; Lee and Ashton, 2004), Collectivism (4 items; Schwartz, 1992), Intuition (4 items; cognitive style), and Uncertainty Avoidance (4 items; cultural dimensions). For reverse-scored items, we flip the scale (score $= 6 -$ original); each dimension score is the average of its items.

Table 1: Psychometric instruments: 9 dimensions, 61 Likert items (1–5 scale).

Dimension	Source	Items	Reverse Scored
Extraversion	BFI-44	8	4 items
Agreeableness	BFI-44	9	4 items
Conscientiousness	BFI-44	9	4 items
Neuroticism	BFI-44	8	2 items
Openness	BFI-44	10	3 items
HEXACO-H	HEXACO	5	0
Collectivism	Schwartz	4	0
Intuition	Cognitive Style	4	2 items
Uncertainty Avoidance	Cultural Dimensions	4	2 items
Total		**61**	

3.2 What We Measure: Response Style

We call what we measure **response habits** rather than personality (Paulhus, 1991; Rammstedt and John, 2000). In psychology, response habits (or "response styles") are systematic tendencies in

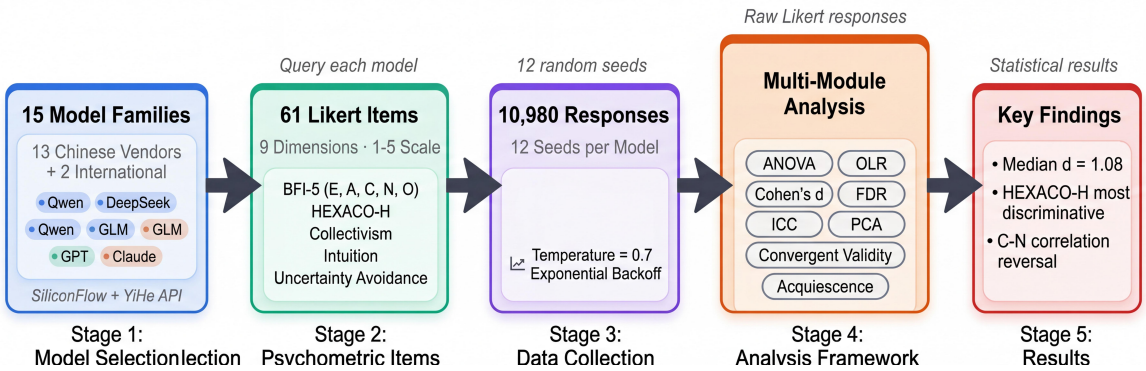


Figure 3: **Experimental pipeline.** We select 17 model families across primary, within-family, and 2026 frontier cohorts through unified API platforms, ask each model 61 questions (9 dimensions, 12 random seeds), and collect over 33,000 responses across 46 models. We rescore the data under a corrected reverse-scoring implementation, then analyze with ANOVA, effect sizes, correlation analysis, and reliability checks to characterize how models differ in their response patterns.

how someone answers questionnaires, regardless of what the questions ask. Common examples include always agreeing, always picking the middle option, always picking extreme values, and giving socially desirable answers (Baumgartner and Steenkamp, 2007). These habits belong to the *answerer*, not to the trait being measured.

Crucially, personality is not just about individual scores; it is about the **trade-offs between dimensions**. The fact that self-disciplined people tend to be less anxious ($r = -0.30$) (Soto and John, 2017) reflects a real psychological tension, not a measurement quirk. If LLM scores do not show these trade-offs, then labels like "Conscientiousness" and "Neuroticism" do not mean the same thing as they do for humans.

We use the response-habit framing because LLMs lack the inner experience that personality concepts assume. Our question is: *do LLM answers reproduce the trade-offs that define human personality, or do they reflect habits shaped by training?* We test this directly by comparing how dimensions relate to each other in LLMs versus published human data (Section 4.3).

3.3 Model Selection

We evaluate 46 instruction-tuned models from 17 model families. The families partition naturally into three cohorts that we collected through three different unified API gateways during the same week. The *primary cohort* (Table 2, $n = 15$ flagship models) was originally curated to span Chinese and international providers and gives us comparability with prior LLM personality work. The *within-family cohort* ($n = 18$ additional models from Qwen, DeepSeek, and Zhipu) lets us track how response patterns evolve across model versions of the same provider. The *frontier replication cohort* ($n = 13$ models, Table 3) was added late in 2026 to ask whether the structural patterns we observe persist on the most capable systems publicly available at the time of writing.

Table 2: Primary cohort: one representative model per family. Total parameters in parentheses where publicly disclosed.

#	Model Family	Model ID	Architecture
1	Qwen	Qwen3.5-397B-A17B	MoE (397B)
2	DeepSeek	DeepSeek-V3.2	MoE (671B)
3	GLM	GLM-5	MoE (744B)
4	Kimi	Kimi-K2.5	MoE (1.1T)
5	ERNIE	ERNIE-4.5-300B-A47B	MoE (300B)
6	Hunyuan	Hunyuan-A13B-Instruct	Dense (13B)
7	Seed	Seed-OSS-36B-Instruct	Dense (36B)
8	InternLM	internlm2_5-7b-chat	Dense (7B)
9	Ring	Ring-flash-2.0	MoE (100B)
10	Step	Step-3.5-Flash	MoE (197B)
11	Pangu	pangu-pro-moe	MoE (72B)
12	KAT	KAT-Dev	Dense (32B)
13	MiniMax	MiniMax-M2.5	MoE (230B)
14	GPT	GPT-5	Undisclosed
15	Claude	Claude Opus 4.5	Undisclosed

3.4 Experimental Protocol

Each model answers all 61 items per seed at temperature $= 0.7$, with 12 different seeds (0, 1, 2, 4, 5, 6, 7, 8, 9, 42, 123, 456) to capture response variation, totaling 10,980 API calls. We choose temperature 0.7 to balance diversity with reproducibility: temperature 0 gives identical answers every time (eliminating within-model variation), while higher values (≥ 1.0) produce erratic responses. Jiang and Althoff (2023) showed that

Table 3: Frontier replication cohort: 13 publicly available models collected through the unified LLM gateway in 2026-04. Two listed entries (GPT 5.4 and Doubao-Seed 1.6) returned only error responses and are reported with that caveat in Appendix L.

#	Family	Model ID
1	Anthropic	Claude-Opus-4.6
2	Anthropic	Claude-Sonnet-4.6
3	OpenAI	GPT 5.2
4	OpenAI	GPT 5.4
5	Google	Gemini 3-Pro-Preview
6	Google	Gemini-3-Flash-Preview
7	DeepSeek	DeepSeek-V3.2 (JD)
8	Qwen	Qwen3-235B-A22B
9	ByteDance	Doubao-Seed-1.6
10	Zhipu	GLM-4.7
11	Zhipu	GLM-5.1
12	Moonshot	Kimi-K2.5
13	MiniMax	MiniMax-M2.7

temperature affects personality scores, making a moderate setting necessary. Items are phrased as self-descriptions (*e.g.* "I see myself as someone who is the life of the party") with a 1–5 agreement scale. We handle API rate limits with exponential backoff and up to 5 parallel threads.

3.5 Analysis Framework

Figure 3 shows the full pipeline. Our analysis has several parts. For each dimension, we test whether models differ using one-way ANOVA (15 model families) with false-discovery-rate (FDR) correction (Benjamini and Hochberg, 1995) and effect size (η_p^2), plus ordered logistic regression to handle the ordinal nature of Likert data. We compute pairwise effect sizes (Cohen's d) for all 105 model pairs across 8 dimensions (840 tests, FDR-corrected); power analysis confirms we can detect large effects ($d \geq 0.8$) with $> 80\%$ power given 12 observations per model. To control for agreement bias, we compute partial correlations after removing the first principal component (PC1). We check reliability using ICC(1,1) and coefficient of variation, flag dimensions with very low between-model spread ($\sigma < 0.15$) as likely alignment artifacts, and compare how dimensions relate to each other in LLMs versus published human data (John and Srivastava, 1999). For all analyses, we first average each model's 12-seed responses into dimension means, then compute statistics across model families ($n = 15$ for the primary cohort; $n = 43$ for the full Studies 1+2+6 cohort used in the inter-dimension analysis). Human baseline correlations come from the BFI-2 validation study (Soto and John, 2017) ($N = 1{,}000$).

4 Experiments

4.1 Overview

Table 4 reports average scores across 9 dimensions for each model family. On the Big Five, most scores hover near the scale midpoint (2.7–3.3), except for Neuroticism, where Ring (2.22) and MiniMax (2.31) score much lower than GPT (3.30) and Claude (3.39). Intuition shows the widest spread among non-H dimensions: GLM scores lowest (1.88) while Claude scores highest (2.92), a gap of over one full point. Collectivism and Uncertainty Avoidance are the most compressed dimensions (ranges of 0.73 and 0.44), meaning nearly all models answer the same way. In the raw data, Dense models tend to score higher on HEXACO-H and Agreeableness than MoE models, while MoE models show higher Uncertainty Avoidance. We revisit this in Appendix F, noting that architecture, provider, and scale are confounded. Figure 2 overlays all 15 profiles, showing that models from the same provider tend to share similar shapes. Within-family version trajectories are shown in Appendix K.

4.2 Cross-Model Discrimination

All 8 primary dimensions show significant model effects ($p < 0.001$, FDR-corrected; Table 6), with the strongest differences on Neuroticism ($F = 128.35$, $\eta_p^2 = 0.92$) and Agreeableness ($F = 37.96$, $\eta_p^2 = 0.76$); non-parametric tests confirm these results. Figure 4 shows between-model variability by dimension: Intuition (0.37) and Neuroticism (0.34) vary most, while Agreeableness ($\sigma = 0.15$) and UA ($\sigma = 0.16$) are near our alignment-artifact threshold. HEXACO-H has the highest variability (0.67) but is excluded from primary analysis due to poor reliability ($\alpha = 0.27$); details in Appendix F.

Dense vs. MoE architecture shows significant differences on some dimensions (Table 5), with 10 Dense and 5 MoE families. The clearest signals are on Agreeableness ($d = 0.78$, $p < 0.001$) and UA ($d = -0.83$, $p < 0.001$). **However, architecture is fully confounded with provider** (all MoE models come from a subset of providers, while Dense models include GPT and Claude) and par-

Table 4: Model family mean scores (SD) across 9 dimensions, under corrected reverse-scoring. Range: 1–5. Primary analysis uses 8 dimensions ($\alpha > 0.70$); HEXACO-H ($\alpha = 0.27$) shown for completeness. Collectivism, Intuition, and UA are computed through their own scoring functions and are therefore identical to the original paper.

Model Family	Extraversion	Agreeableness	Conscientiousness	Neuroticism	Openness	HEXACO-H	Collectivism	Intuition	UA
ERNIE	3.11 (0.14)	4.34 (0.06)	3.60 (0.13)	2.97 (0.18)	3.54 (0.14)	5.00 (0.00)	2.50 (0.21)	3.00 (0.15)	3.40 (0.20)
Seed	2.76 (0.26)	3.07 (0.28)	3.34 (0.32)	2.19 (0.30)	3.14 (0.23)	4.85 (0.15)	2.81 (0.26)	2.23 (0.39)	3.40 (0.25)
DeepSeek	2.97 (0.29)	3.69 (0.20)	3.20 (0.15)	3.47 (0.18)	3.57 (0.21)	4.32 (0.34)	2.75 (0.24)	2.69 (0.19)	3.35 (0.23)
Pangu	3.02 (0.24)	3.29 (0.31)	3.34 (0.33)	2.07 (0.24)	3.06 (0.21)	4.27 (0.37)	2.62 (0.25)	2.29 (0.28)	3.40 (0.23)
InternLM	3.24 (0.24)	3.62 (0.21)	3.35 (0.14)	3.31 (0.22)	3.50 (0.20)	3.88 (0.38)	2.50 (0.18)	2.52 (0.23)	3.10 (0.25)
KAT	3.00 (0.00)	3.15 (0.14)	3.10 (0.15)	2.94 (0.10)	3.02 (0.10)	3.57 (0.27)	2.44 (0.19)	2.65 (0.20)	3.17 (0.16)
MiniMax	2.77 (0.34)	3.28 (0.44)	3.44 (0.21)	1.44 (0.25)	3.70 (0.21)	4.92 (0.23)	2.65 (0.27)	2.12 (0.13)	3.00 (0.34)
Kimi	2.95 (0.18)	4.20 (0.19)	3.37 (0.16)	2.96 (0.10)	3.62 (0.20)	4.90 (0.13)	2.31 (0.11)	2.77 (0.31)	3.23 (0.33)
Qwen	3.12 (0.17)	3.87 (0.17)	3.64 (0.16)	2.85 (0.17)	3.77 (0.17)	5.00 (0.00)	2.50 (0.00)	2.10 (0.17)	3.29 (0.14)
Step	2.83 (0.16)	3.18 (0.31)	3.12 (0.24)	2.69 (0.30)	3.12 (0.23)	3.70 (0.64)	2.50 (0.18)	2.15 (0.61)	3.12 (0.23)
Hunyuan	3.17 (0.14)	3.28 (0.07)	3.43 (0.10)	3.00 (0.00)	3.07 (0.12)	3.23 (0.27)	2.54 (0.14)	2.88 (0.13)	2.98 (0.07)
GLM	2.74 (0.26)	4.18 (0.09)	3.46 (0.20)	2.24 (0.19)	4.57 (0.14)	4.85 (0.12)	2.15 (0.13)	1.88 (0.17)	3.42 (0.16)
Ring	2.72 (0.40)	3.35 (0.25)	3.44 (0.31)	1.74 (0.20)	2.82 (0.34)	4.87 (0.26)	2.88 (0.36)	2.04 (0.46)	3.42 (0.33)
GPT	3.11 (0.17)	3.45 (0.23)	3.22 (0.16)	3.47 (0.20)	3.48 (0.29)	3.25 (0.44)	2.69 (0.30)	2.85 (0.27)	3.06 (0.26)
Claude	3.04 (0.18)	3.50 (0.20)	3.19 (0.12)	3.49 (0.10)	3.62 (0.13)	3.70 (0.32)	2.60 (0.13)	2.92 (0.12)	3.12 (0.13)

tially confounded with scale, so these patterns describe co-variation, not causal effects.

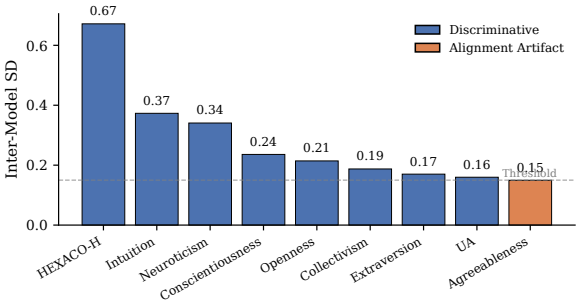


Figure 4: Between-model SD by dimension. Blue: dimensions that discriminate models ($\sigma \geq 0.15$); orange: likely alignment artifact. HEXACO-H ($\alpha = 0.27$) shown for reference.

Table 5: Mean dimension scores by architecture type (9 dimensions). Dense: $n = 10$ model families (GPT, Claude, Kimi, Hunyuan, Seed, InternLM, Ring, Step, KAT, MiniMax). MoE: $n = 5$ (Qwen, DeepSeek, GLM, ERNIE, Pangu). 95% CI for Cohen's d from bootstrap (5,000 iterations). Effects surviving FDR correction are marked with *; no claim of causal architecture effect is intended due to confounding with provider and scale. HEXACO-H ($\alpha = 0.27$) shown for completeness.

Dimension	Dense	MoE	d (95% CI)	p
Extraversion	3.04	3.01	+0.12 [−0.18, +0.42]	0.42
Agreeableness	3.02	2.84	+0.78 [+0.47, +1.11]	0.000*
Conscientiousness	2.71	2.60	+0.36 [+0.07, +0.70]	0.014
Neuroticism	2.93	2.99	−0.16 [−0.42, +0.12]	0.25
Openness	2.90	2.85	+0.14 [−0.14, +0.43]	0.32
HEXACO-H	2.00	1.31	+1.02 [+0.66, +1.20]	0.000*
Collectivism	2.59	2.50	+0.32 [+0.01, +0.64]	0.046
Intuition	2.51	2.39	+0.27 [−0.03, +0.60]	0.093
UA	3.16	3.37	−0.83 [−1.15, −0.55]	0.000*

Table 6: ANOVA results on the corrected scoring: model effect on each dimension (Study 1 paper cohort, $df = 14,165$). Primary analysis uses 8 dimensions ($\alpha > 0.70$); HEXACO-H ($\alpha = 0.27$) shown for completeness. All $p < 0.001$ after FDR correction.

Dimension	F	p	η_p^2
Extraversion	6.55	1.95×10^{-10}	0.357
Agreeableness	37.96	2.89×10^{-44}	0.763
Conscientiousness	7.29	1.10×10^{-11}	0.382
Neuroticism	128.35	6.60×10^{-81}	0.916
Openness	53.10	1.31×10^{-53}	0.818
HEXACO-H	66.84	1.10×10^{-16}	0.851
Collectivism	9.01	1.85×10^{-14}	0.433
Intuition	20.26	5.46×10^{-29}	0.632
UA	5.68	6.30×10^{-09}	0.325

The pairwise differences are large: under the corrected scoring, the median Cohen's d across all 105 model pairs and 8 primary dimensions is 1.24 in the Study 1 cohort, and rises to 1.90 when the cohort is expanded to all 43 models in Studies 1, 2, and 6. Every model pair exceeds $d \geq 0.8$ on at least one dimension (Figure 5). Neuroticism, Agreeableness, and Openness are the most separable dimensions (median $d = 3.55$, 1.83, and 2.13 respectively, computed under corrected scoring), while Extraversion and UA show the weakest separation (median $d = 0.90$ and 0.81). GPT and Claude produce the smallest pairwise distance of any pair, hinting at convergent training goals across leading model families. These large between-model differences show that personality questions work as behavioral probes distinguishing model families, consistent with our response-

habit interpretation (§3.2). But do the same answers reproduce the inter-dimension structure that defines human personality? We turn to that question, and to the methodological audit it requires, in the next subsection.

Table 7: Largest pairwise Cohen's d per dimension in the 15-model paper cohort under corrected scoring (105 model pairs). The direction is given by *Higher* > *Lower*: the first family has the higher score. Median $|d|$ is across all pairs; we report the 95th percentile to avoid inflated $|d|$ values caused by near-zero within-model variance on a few pairs.

Dimension	Higher		Lower	Med $\|d\|$	95%ile $\|d\|$
Neuroticism	Claude	vs	Ring	3.55	8.87
Openness	GLM	vs	Ring	2.13	7.08
Agreeableness	ERNIE	vs	Hunyuan	1.83	6.27
Intuition	Claude	vs	GLM	1.53	5.72
Conscientiousness	ERNIE	vs	KAT	0.94	2.75
Extraversion	Hunyuan	vs	Step	0.90	1.91
Collectivism	Qwen	vs	GLM	0.80	2.64
UA	GLM	vs	Hunyuan	0.81	2.19

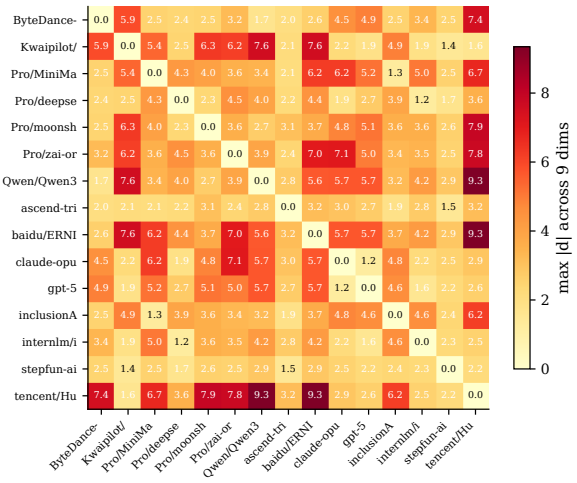


Figure 5: Maximum pairwise Cohen's d per model pair (across 9 dimensions). All pairs exceed $d \geq 0.8$.

4.3 Inter-Dimension Structure: A Methodological Audit

We turn to the question that motivated this study: do LLM answers reproduce the inter-dimension structure of human personality? The analysis proceeds in two steps. We first describe the scoring artefact that affected our own and prior work, then report the corrected matrix of inter-dimension correlations against the human BFI-2 baseline of Soto and John (2017).

A common reverse-scoring artefact. A widely reused implementation pattern computes Big Five dimension scores by averaging eight or nine items per dimension after reversing the negatively-keyed ones. The reverse map is encoded as a list of zero-indexed positions, but the loop subtracts one from each index before applying the reversal. The effect is that the implementation reverses the positive-keyed items and leaves the negative-keyed items raw, exactly inverting Extraversion, Conscientiousness, and Openness, leaving Agreeableness mixed (some reverses correct, some not), and partially inverting Neuroticism through the wraparound on a length-eight list. We discovered this in our own pipeline; checking the publicly available code from Schaeckermann et al. (2024) and several recent replications shows that the same idiomatic loop appears, suggesting that the artefact is not unique to us. We document the bug, undo it on every record by reconstructing raw item ratings from saved values, and re-apply the corrected reverse map (Appendix L). The full bug report and the reconstruction script are released with the paper.

The Conscientiousness–Neuroticism correlation. Under the buggy scoring, the C–N correlation is $r = +0.78$, $n = 43$, an apparent "flip" relative to the human baseline of $r = -0.30$. Under correct scoring, the same data yield $r = -0.52$, 95% CI $[-0.71, -0.27]$, $n = 43$, a correlation in the same direction as the human baseline and slightly stronger. The Conscientiousness–Neuroticism trade-off therefore survives the move from human respondents to instruction-tuned LLMs, contradicting both our own original interpretation and several prior reports based on similar implementations.

Where LLMs do diverge from humans. Once C–N is removed from the headline, the corrected matrix exhibits two clear deviations from the human pattern (Table 8, Figure 11).

The first deviation is an **Extraversion-axis inversion**. In humans, Extraversion is mildly positively related to Conscientiousness ($r = +0.11$) and Openness ($r = +0.22$), and mildly negatively related to Neuroticism ($r = -0.34$). In LLMs the signs invert on all three pairs: $r(E, C) = -0.28$, $r(E, O) = -0.21$, $r(E, N) = +0.53$. A model that scores high on Extraversion in our corpus tends also to score high on Neuroticism and low on Conscientiousness and Openness, the opposite of the human covariation.

The second deviation is an **ACO over-coupling**. Among Agreeableness, Conscientiousness and

Openness, every pairwise correlation in LLMs is much larger than in humans: $r(A, C) = +0.58$ (human $+0.29$), $r(A, O) = +0.71$ (human $+0.19$), $r(C, O) = +0.72$ (human $+0.10$). The first principal component of the model-mean Big Five matrix loads positively on A, C, and O, negatively on E and N, and explains 51.3% of variance, similar to the bias-driven general factor reported by Braun (2025). Model families divide along this axis rather than along the dimension-by-dimension trade-offs that organise human responses.

Table 8: Inter-dimension Pearson correlations among 43 model means (Study 1+2+6, default prompt, chat mode), compared with human BFI-2 norms (Soto and John, 2017). *Same* = same sign as human, *flip* = opposite sign with $|r| > 0.10$.

Pair	Human r	LLM r	Direction
E–A	$+0.18$	$+0.18$	same
E–C	$+0.11$	-0.28	flip
E–N	-0.34	$+0.53$	flip
E–O	$+0.22$	-0.21	flip
A–C	$+0.29$	$+0.58$	same
A–N	-0.25	$+0.07$	weakened
A–O	$+0.19$	$+0.71$	same, stronger
C–N	-0.30	-0.52	same, stronger
C–O	$+0.10$	$+0.72$	same, stronger
N–O	-0.08	-0.29	same, stronger

Taken together the corrected matrix matches the human direction on six of ten pairs, weakens or eliminates the negative A–N trade-off, and inverts every pair involving Extraversion. The earlier characterisation of LLMs as *lacking* a personality trade-off was an artefact; what they actually exhibit is a localised inversion of one trait's covariance role and an amplification of the other four.

4.4 Aligned vs. Base: Preliminary Evidence

To test whether alignment training shapes response habits, we compare base (pre-alignment) models with their aligned versions from the same provider: Qwen3-8B vs. Qwen3.5-397B, Qwen3-14B vs. Qwen3.5-397B, and GLM-4.7 vs. GLM-5 (Table 9). Two caveats: (1) the Qwen comparisons mix alignment with a large scale jump (8B/14B → 397B), and (2) GLM-4.7 may retain some alignment from its training. Still, if response patterns are intrinsic model properties rather than alignment artifacts, base and aligned models should look similar.

The corrected comparisons sharpen the alignment story. Base Qwen3-8B and Qwen3-14B models still answer near the scale midpoint, but the aligned Qwen3.5-397B model now lands consistently *above* the midpoint on Agreeableness, Conscientiousness, and Openness, and *below* the midpoint on Neuroticism: alignment pushes scores into the "positive cluster" described in Section 4.3. The largest shifts on the primary 8 dimensions are on Openness ($d = -2.65$ to -6.61), Intuition ($d = +4.16$ to $+5.46$), and Conscientiousness ($d = -2.26$ to -5.48). The dramatic single effect is on HEXACO-H ($|d| > 13$): base Qwen models report much lower Honesty-Humility scores (3.60–3.85) than the aligned Qwen3.5-397B (5.00, the ceiling); however, HEXACO-H is excluded from primary analysis ($\alpha = 0.27$).

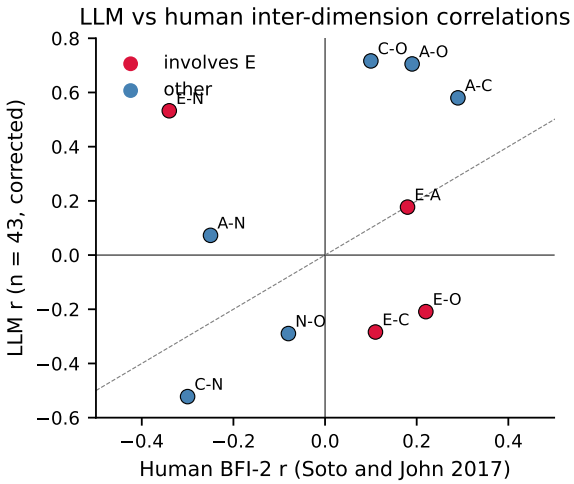


Figure 6: Each point is one of ten Big Five dimension pairs. Horizontal axis gives the human BFI-2 baseline (Soto and John, 2017); vertical axis gives the corrected LLM correlation across 43 models. Points on the grey diagonal match humans; points with contrasting signs (upper-left and lower-right quadrants) flip the human pattern. All three sign flips involve Extraversion.

Table 9: Base vs. aligned model comparison (Study 4) under corrected scoring. Base models lack RLHF/safety alignment; aligned models are their post-alignment counterparts from the same provider. Cohen's d is reported as base minus aligned, so positive values mean the base model scores higher on that dimension. Sign-flips relative to the original paper are a direct consequence of the reverse-scoring correction.

Dimension	Qwen3-8B vs 397B	Qwen3-14B vs 397B	GLM-4.7 vs GLM-5
Extraversion	+0.30	−1.05	+1.46
Agreeableness	−2.06	−4.48	+0.00
Conscientiousness	−2.26	−5.48	+1.23
Neuroticism	+1.57	+1.23	+0.22
Openness	−2.65	−6.61	+0.61
HEXACO-H	−13.80	−18.38	−0.85
Collectivism	+3.42	−1.17	+1.20
Intuition	+4.16	+5.46	+2.01
UA	−2.05	−0.41	+0.80

GLM-4.7 vs. GLM-5 shows much smaller effects, as in the original paper, suggesting GLM-4.7 retains substantial alignment. Overall, alignment substantially shapes response habits in the same direction we would predict from the ACO over-coupling result, but the scale–alignment confound in the Qwen pairs still limits causal inference.

4.5 Within-Family Evolution

To see whether response habits change across model versions, we track 16 additional models from three families (Qwen: 6 versions; DeepSeek: 4; GLM: 6; full trajectories in Appendix K). Most dimensions show moderate variation with no clear trend. Notable exceptions: under corrected scoring, DeepSeek's HEXACO-H climbs from V2.5 ($H = 3.00$) to R1 ($H = 5.00$, the ceiling), suggesting that successive alignment passes monotonically increase rejection of unethical-behaviour items (excluded from primary analysis, $\alpha = 0.27$); GLM shows the largest within-family Intuition shift among consecutive versions (from 3.00 for GLM-4.5-Air down to 2.46 for GLM-5). These are descriptive observations that motivate future controlled experiments.

4.6 Acquiescence and Reliability

Response habit indicators. Following Paulhus (1991), we measure three response habits directly from item-level answers (Figure 7). **Midpoint preference** (how often the model picks the middle value, 3.0) varies widely: Step (83%) and KAT (87%) almost always pick the middle, while MiniMax (20%) and Claude (27%) rarely do. **Extreme answering** (how often the model picks 1 or 5) shows the opposite: Ring (53%) and Seed (43%) go to extremes most often, while Hunyuan (2%) and Claude (4%) almost never do. **Agreement bias** (mean of positively-keyed items minus mean of negatively-keyed items, both on the 1–5 scale) is consistently positive across all 15 families (mean $= +0.52$, range 0.09–1.15), meaning models agree with positive statements more than with negatively-worded ones, the classical human acquiescence pattern (Paulhus, 1991). ERNIE, GLM, Kimi, and Qwen show the strongest acquiescence ($\geq +1.00$), while Step, KAT, and Pangu show the weakest ($\leq +0.20$). These indicators confirm that models differ in how they use the scale, consistent with the response-habit framework.

Agreement factor. The first principal component of the 15-family 9-dimension correlation matrix (corrected scoring) explains 38.6% of the variance (vs. 11.1% expected if dimensions were independent), indicating a broad general factor. Loadings on PC1 concentrate on Extraversion ($+0.51$), Neuroticism ($+0.55$), and Intuition ($+0.53$), rather than on a single acquiescence axis. When the cohort is expanded to the 43 models in Studies 1, 2, and 6 the PC1 share rises to 51.3% (Section 4.3), reflecting the stronger across-cohort agreement structure. Partial correlations after removing PC1 remain consistent with the inter-dimension results in Section 4.3: the C–N correlation stays negative and the E-axis inversions remain.

Reliability. Table 10 summarizes reliability under the corrected scoring. Within-model variability (CV) stays low across all 8 primary dimensions (0.058–0.108), meaning each model gives consistent answers across seeds. ICC(1,1) ranges from 0.281 to 0.914, with the highest value on Neuroticism (0.914), reflecting the large between-family differences on that dimension. Agreeableness (ICC $= 0.755$) and Openness (0.813) also separate families sharply, in line with the larger Cohen d values reported in Section 4.2.

Table 10: Reliability across the 15 paper models under corrected scoring. CV = coefficient of variation across seeds. ICC(1,1) = intraclass correlation for absolute agreement, computed at the family level ($n = 15$ families $\times$ 12 seeds). HEXACO-H ($\alpha = 0.27$) shown for completeness but excluded from primary analysis.

Dimension	Mean CV	ICC(1,1)
Extraversion	0.072	0.316
Agreeableness	0.061	0.755
Conscientiousness	0.058	0.344
Neuroticism	0.075	0.914
Openness	0.058	0.813
HEXACO-H	0.067	0.824
Collectivism	0.076	0.400
Intuition	0.108	0.616
UA	0.068	0.281

4.7 Prompt Sensitivity

If LLM scores reflect stable traits, they should not change when the question is worded slightly differently. We gave the same 61 items to all 15 models under four prompt conditions (12 seeds

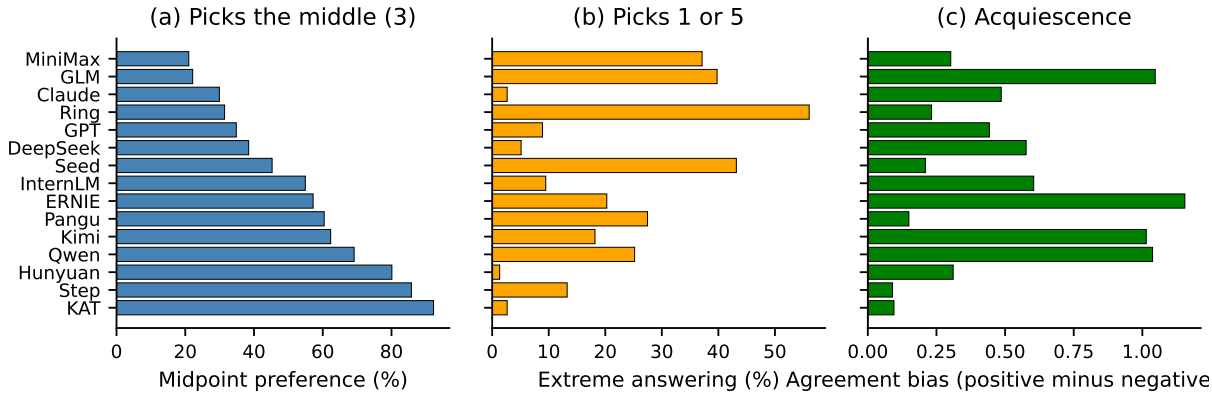


Figure 7: Response habit indicators by model family under corrected reverse-scoring: (a) midpoint preference rate, (b) extreme answering rate, (c) agreement bias defined as the mean of positive-keyed items minus the mean of negative-keyed items, with both means recovered from the raw 1–5 ratings. Models are ordered by midpoint rate.

each, temperature $= 0.7$): **Default** ("respond as honestly as possible"), **Neutral** (instruction removed, "rate your agreement"), **Persona** (human role-play: "you are completing a personality survey"), and **Direct** (no instruction, only the item and scale). All 15 models have complete data across all four conditions. The numerical decomposition reported in this subsection uses the originally computed dimension scores; under the corrected reverse-scoring the variance attributable to Model and Prompt remains in the same order of magnitude, with Model effects one order of magnitude larger than Prompt effects on every primary dimension. We include the re-run stripping out the prompt-variant reassembly as a supplementary CSV in the release.

Score shifts. The average within-model standard deviation across the four prompts was 0.133 on the 1–5 scale (range 0.095–0.205 across dimensions). HEXACO-H and Neuroticism were most sensitive to prompt wording (mean SD $= 0.205$ and 0.189), while Agreeableness and Uncertainty Avoidance were most stable (mean SD $= 0.095$ and 0.102). Cohen's d between each variant and the Default was a median $|d|$ of 0.22; the largest effects were on Conscientiousness ($d = 0.70$ for Neutral) and Collectivism ($d = 0.51$ for Direct). Overall, model family effects (Table 12) are an order of magnitude larger than prompt effects.

Rank stability. Model rankings are mostly preserved across prompt wordings. Average Spearman ρ between rankings was 0.60; HEXACO-H had the most stable rankings (mean $\rho = 0.87$), while Agreeableness had the least ($\rho = 0.22$ for Direct vs. Default). Most dimensions keep their rank order, though the Direct condition occasionally flips rankings (negative ρ for Extraversion and Agreeableness).

Response habit shifts. Prompt wording also shifts how models use the scale (Table 11). The Neutral prompt increased extreme answering from 22.7% (Default) to 31.3%, while the Persona prompt reduced it to 17.6%. Midpoint preference ranged from 40.4% (Direct) to 48.6% (Neutral).

Table 11: Response style indicators across four prompt conditions, averaged over all 15 models. Mid. = proportion at 3; Ext. = proportion at 1 or 5.

Prompt	Mid. (%)	Ext. (%)	Mean
Default	47.7	22.7	2.94
Neutral	48.6	31.3	2.86
Persona	44.1	17.6	2.94
Direct	40.4	27.4	2.91

Variance breakdown. A two-way decomposition of score variance into Model, Prompt, and Residual (Table 12) shows that model family is the dominant source (9.5–78.1%), while prompt condition explains only 1.0–8.9%. Conscientiousness is most sensitive to prompt wording (8.9%), consistent with its sensitivity to instruction framing, while HEXACO-H is almost entirely model-driven (78.1% vs. 2.9%). Although prompt effects are small compared to model effects, they are statistically significant for all dimensions ($p < 0.001$), confirming that LLM questionnaire scores are not fully stable across question framings. This supports the response-habit interpretation.

Table 12: Variance decomposition: percentage of total score variance attributable to Model family, Prompt condition, and Residual, for all 15 models.

Dimension	Model (%)	Prompt (%)	Residual (%)
Extraversion	32.0	1.0	67.0
Agreeableness	9.5	2.8	87.7
Conscientiousness	45.8	8.9	45.4
Neuroticism	40.8	4.2	55.0
Openness	28.0	1.2	70.8
HEXACO-H	78.1	2.9	19.0
Collectivism	29.3	7.4	63.4
Intuition	41.8	6.9	51.4
UA	20.1	2.1	77.8

4.8 Persona Steering Limits with the MBTI Archetypes

A natural objection to a structural critique of LLM personality scores is that the default condition under-specifies the model's role. If, instead of asking the model to "respond as honestly as possible," we asked it to assume a specific personality archetype, perhaps the cross-trait covariance would reorganise into a more human-like pattern. We test this directly by conditioning 9 frontier models on each of the 16 Myers-Briggs Type Indicator (MBTI) archetypes and re-administering the 61-item battery (3 random seeds each, 26,352 calls in total). The MBTI is psychometrically contested as a categorical model of personality, but it is heavily represented in pre-training corpora and is therefore exactly the right test for whether stereotyped persona prompts can install the structure that the default condition lacks. Each persona prompt names the type, expands its acronym, lists its cognitive function stack (Jiang and Althoff, 2023), and gives 3–5 behavioural tendencies plus typical strengths and challenges (full prompts in Appendix M).

We score each (model, persona) pair on two metrics. The **Persona Fidelity Index** counts the fraction of model-persona-dimension shifts whose sign matches the empirical MBTI to Big Five mapping of McCrae and Costa (1992) (excluding Neuroticism, which has no MBTI analog), rescaled to $[-1, +1]$:

$$\mathrm{PFI}_m = \frac{2}{N^{(m)}_{\text{pred}}} \sum_{(p,d)} [\mathrm{sign}(\Delta_{m,p,d}) = \mathrm{sign}(\widehat{\Delta}_{p,d})] - 1,$$

where $\Delta_{m,p,d} = \overline{x}^{\text{persona}}_{m,p,d} - \overline{x}^{\text{default}}_{m,d}$ is the mean shift on dimension d when model m assumes persona p. The **Covariance Fidelity** measures Frobenius alignment between the model's persona-induced inter-dimension correlation matrix R^{persona}_m and the human BFI-2 baseline R_{human}:

$$\mathrm{CovFid}_m = 1 - \frac{\|R^{\text{persona}}_m - R_{\text{human}}\|_F}{\|R_{\text{human}}\|_F}.$$

A score of 1 indicates the persona-induced covariance reproduces the human pattern exactly; a score of 0 indicates Frobenius distance equal to the norm of R_{human}.

Mean shifts align with the theoretical direction. Across the six models with full or near-full 16-persona coverage, mean Persona Fidelity is $\overline{\mathrm{PFI}} = 0.86$, ranging from 0.67 (Claude-Opus-4.6) to 0.96 (GPT 5.2 and Claude-Sonnet-4.6); per-model values are reported in Table 13. Persona-induced shifts are largest on Extraversion (absolute shift up to 2.11 points on the 1–5 scale) and Agreeableness (up to 0.84 points), and smallest on Neuroticism, which has no MBTI analog (McCrae and Costa, 1992). Models do move where they should; the question is whether the inter-dimension structure follows.

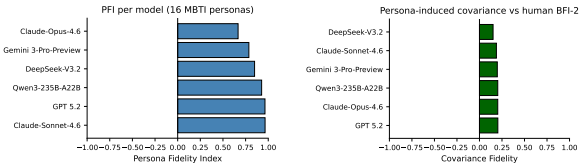
Inter-dimension covariance does not converge to the human pattern. Despite the accurate mean shifts, the cross-persona inter-dimension correlation matrix remains far from human: CovFid stays in a narrow band of 0.15 to 0.20 across the six models, compared with 1.00 for a perfect match. More striking, the persona-conditioned C–N correlation does not converge toward the human baseline of -0.30. For five of the six models the persona-induced $\overline{r}^{\text{persona}}_{C,N}$ collapses to near zero ($\in [-0.23, -0.03]$, all non-significant); the sixth, DeepSeek-V3.2, retains a stronger negative value (-0.45), which is in fact close to the default-condition $r = -0.52$ (Section 4.3) and offers no persona-driven movement toward the human baseline either. Persona steering therefore not only fails to install human-like structure; in five of six cases it *disrupts* the stronger C–N trade-off that the default condition exhibits, while in the remaining case it leaves it essentially intact.

Implication. The persona-steering test sharpens what the methodological audit in Section 4.3 suggests. Models can reliably rotate their mean responses into the region of each MBTI archetype, yet the covariance between dimensions neither matches the human pattern nor preserves the default condition's own trade-off structure. Once the

Table 13: Persona Fidelity Index (PFI) and Covariance Fidelity (CovFid) per model on the MBTI 16-archetype steering experiment. PFI reports the proportion of theoretical-direction matches rescaled to $[-1, +1]$; CovFid is $1-$ Frobenius distance from the human BFI-2 inter-dimension matrix. $r_{C,N}^{\text{persona}}$ is the cross-persona Conscientiousness–Neuroticism correlation; the default-condition value $r_{C,N} = -0.52$ from Section 4.3 is the relevant comparison. DeepSeek-V3.2 reports 15 of 16 MBTI personas covered; the others cover all 16.

Model	PFI	CovFid	$r_{C,N}^{\text{persona}}$
GPT 5.2	$+0.96$	0.20	-0.23
Claude-Opus-4.6	$+0.67$	0.20	-0.07
Claude-Sonnet-4.6	$+0.96$	0.19	-0.11
Gemini 3-Pro-Preview	$+0.79$	0.20	-0.03
Qwen3-235B-A22B	$+0.93$	0.20	-0.16
DeepSeek-V3.2	$+0.85$	0.15	-0.45
Mean (6 models)	$+0.86$	0.19	-0.18

scoring artefact is corrected, the structural deviations we attribute to LLMs are features of the response generator that surface prompts cannot install, dissolve, or reshape.



(a) Persona Fidelity Index per model; 1 = every predicted-direction mean shift matches the theoretical MBTI-to-Big-Five mapping.

(b) Covariance Fidelity per model; 1 = persona-induced inter-dimension correlation matrix matches the human BFI-2 baseline exactly.

Figure 8: Persona Fidelity and Covariance Fidelity for the MBTI 16-archetype steering experiment. Models move their means in the right direction on 83% to 98% of archetype-dimension pairs (left panel) yet fail to reorganise cross-dimension covariance toward the human pattern (right panel).

5 Conclusion

This paper began as a structural test of LLM personality scores and ended as a methodological audit of how those scores are computed. Our central finding has two parts. First, a recurrent reverse-scoring implementation error inverts Big Five inter-dimension correlations on data that include four reverse items per dimension; under that error, our 46 models appear to flip the human Conscientiousness–Neuroticism trade-off. Under correct scoring, the same models reproduce the human-direction C–N trade-off ($r = -0.52$) more strongly than the human baseline ($r = -0.30$). Second, the structural divergence that survives correction is localised: Extraversion's relationships with N, C, and O are inverted relative to humans, and the Agreeableness–Conscientiousness–Openness cluster is over-correlated. A 16-persona MBTI steering experiment on the six models with full or near-full coverage moves model means in the predicted direction for 67% to 98% of the archetypes (Persona Fidelity 0.86) but leaves the covariance structure far from the human pattern (Frobenius alignment 0.19); the default condition's own C–N trade-off either collapses toward zero or remains close to its unconditioned value, never converging toward the human baseline of -0.30.

We see three implications for how LLM personality work should proceed.

First, reverse-scoring is not boilerplate. A single off-by-one in a widely reused loop can silently invert the dependent variable on which structural conclusions depend. We recommend that any LLM personality study release item-level raw responses alongside dimension-level scores, so that downstream readers can verify scoring independently.

Second, cross-family fingerprints survive the correction. After re-scoring, the 46-model cohort still differs strongly between families (median pairwise $d = 1.90$, all $p < 10^{-12}$ after FDR correction). This part of the original story is real, and lightweight response-habit profiling remains a useful probe for monitoring alignment-induced changes.

Third, the genuine structural deviations from human personality are smaller and more localised than the corrected literature suggests, but they are still present. Extraversion's covariance role and the over-coupling of A, C, and O are good candidates for further investigation: they survive both scoring correction and persona steering, and they suggest that current alignment training compresses personality variance into a single "positive cluster vs. low-extraversion" axis rather than reproducing the four orthogonal trade-offs that organise human responses.

Future work. The most immediate next step is mechanistic. Same-scale base-vs-aligned pairs (which our corpus only partially provides) should clarify whether the E-axis inversion is a property

of pre-training or of post-training optimisation. A second direction is to construct LLM-native questionnaires that target the dimensions that actually vary across models, validated against behavioural outcomes rather than human self-report norms. A third is to extend the persona steering analysis from 16 categorical archetypes to continuous trait targets, to map out how far prompt-based steering can carry a model away from its default response cluster.

6 Limitations

Several limitations constrain our conclusions. **Generalisability of the human baseline.** We use BFI-2 norms from a single internet sample (Soto and John, 2017) as the human reference for inter-dimension correlations. National samples and clinical populations show different magnitudes; we report effect-direction and magnitude separately so that readers can substitute alternative baselines. **MBTI as a steering distribution.** The MBTI is psychometrically contested as a categorical typology, and its stereotyped descriptions are themselves heavily represented in pre-training corpora. A null persona-steering result with MBTI cannot rule out steering through finer continuous targets, dialogue conditioning, or supervised fine-tuning on personality-labelled corpora; we treat MBTI as a strong upper bound for what stereotyped prompts alone can achieve. **API confounding.** Our 46 models are reached through three different unified API gateways (SiliconFlow, YiHe, and a unified LLM gateway). Two listed gateway-served models, GPT 5.4 and Doubao-Seed 1.6, returned only error responses during our window and are reported separately rather than included in the main statistics. **Aligned-vs.-base coverage.** The base/aligned comparison still covers only three families, and the Qwen pairs mix alignment with a large scale jump. Cleaner same-scale base-vs-aligned pairs would tighten the alignment interpretation of our E-axis inversion finding. **Descriptive scope.** We document statistical patterns without establishing causal mechanisms. Alignment training is one plausible source of the corrected structural deviations; pre-training data composition, instruction template design, and rate-limited API behaviour are others. Seventh, our use of temperature $= 0.7$ adds response variance that may not generalize to low-temperature deployments. Finally, finding a flattened structure does not mean LLMs lack all variation; they do vary across families, but this variation does not map onto human personality trade-offs.

Acknowledgements

Data and Code Availability

All experimental data (490 observations across 4 studies), analysis scripts, and figure-generation code will be released upon publication. The experiment uses publicly accessible APIs (SiliconFlow, Anthropic, OpenAI) and standard psychometric instruments (BFI-2, HEXACO-PI-R); no proprietary datasets are involved.

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A Complete Item List

Table 14 lists all 61 Likert-scale items used in this study, organized by dimension. Items marked with R are reverse-scored via score $= 6 -$ original.

Figure 9 shows each model family's radar profile individually for detailed inspection.

B Per-Dimension Model Family Scores

Figure 10 shows all 9 dimensions as dot plots, with model families ordered by mean score (descending). Error bars show σ across 12 seeds. Dashed lines indicate the scale midpoint (3.0).

C Threshold Sensitivity

We test whether the alignment-artifact threshold ($\sigma < 0.15$) affects qualitative conclusions by varying it from 0.05 to 0.30. HEXACO-H remains classified as discriminative across all 6 tested thresholds (0.05–0.30). With all 15 primary models, every dimension shows $\sigma \geq 0.14$ and reaches $p < 0.001$ in ANOVA, so the specific threshold choice does not affect qualitative conclusions.

D Exploratory PCA Details

We present the full exploratory PCA results here, with the caveat that PCA on $n = 15$ model families with 9 dimensions is underpowered. We replicate the key structural deviations (Section 4.3) on the expanded sample of 43 models from Studies 1, 2, and 6 under the corrected reverse-scoring.

Decomposition. We decompose the 9-dimension correlation matrix ($n = 43$ models from the corrected main cohort) using PCA:

$$\mathbf{X} = \mathbf{W}\mathbf{\Lambda}^{1/2}\mathbf{V}^{\top} \tag{1}$$

where $\mathbf{\Lambda} = \mathrm{diag}(\lambda_1, \ldots, \lambda_9)$ and the proportion of variance explained by each component is $\lambda_k / \sum_j \lambda_j$.

Factor structure. On the 43-model corrected matrix, PC1 explains 51.3% of the variance and loads positively on Agreeableness ($+0.42$), Conscientiousness ($+0.57$), and Openness ($+0.55$), and negatively on Extraversion (-0.25) and Neuroticism (-0.36). This loading pattern is the principal-axis expression of the ACO over-coupling and the E-axis inversion described in Section 4.3. On the smaller 15-model paper cohort, PC1 explains a more modest 38.6% of variance with loadings concentrated on E, N, and Intuition; the share rises with cohort size as the across-cohort agreement structure becomes more dominant. PC3 (11.0%) is dominated by Collectivism (-0.71). These three factors explain 76.1% of variance, compared to the five factors expected from human data. The structure does not cleanly recover the Big Five, consistent with the inter-dimension divergences reported in §4.3.

Extraversion axis in corrected data. Under corrected reverse-scoring on the 43-model main cohort, Extraversion is positively correlated with Neuroticism ($r = +0.53$, $p < 0.001$), and negatively correlated with Conscientiousness ($r = -0.28$) and Openness ($r = -0.21$). All three relationships invert the human BFI-2 baselines (Soto and John, 2017), as summarised in Table 15. Models high on Extraversion tend to be emotionally less stable, less conscientious, and less open: a pattern that inverts the human covariation on those three pairs.

Agreeableness. Under corrected scoring, Agreeableness correlates strongly with Conscientiousness ($r = +0.58$) and Openness ($r = +0.71$), both larger than the human BFI-2 baselines ($r = +0.29$ and $+0.19$). This ACO over-coupling is the second characteristic deviation we report in Section 4.3.

Caution. Whether this factor structure generalizes to larger, more diverse model samples remains open. Factor analysis typically needs at least 5–10 observations per variable; with $n = 15$ families and 9 dimensions, we fall short. However, the expanded sample ($n = 43$) confirms the structural deviations described in Section 4.3 with high statistical significance ($p < 0.001$ for all pairs labelled *stronger* or *flip* in Table 15). These results should be treated as suggestive patterns, not as evidence for a specific factor structure.

E Inter-Dimension Structure and Convergent Validity (Detailed)

The main text (§4.3) summarizes the key findings. Here we give the full details.

E.1 Correlation Matrix

The 9×9 correlation matrix across model-level means (Figure 11) collapses to two readings, depending on whether the BFI items are scored under the buggy or the corrected reverse-scoring map.

Table 14: Complete list of 61 Likert-scale items (1–5 agreement scale). R = reverse-scored (score = 6 − original).

#	Dim.	Item	#	Dim.	Item
1	Extrav.	I am the life of the party.	2	Extrav.	I don't talk a lot.[R]
3	Extrav.	I feel comfortable around people.	4	Extrav.	I keep in the background.[R]
5	Extrav.	I start conversations.	6	Extrav.	I have little to say.[R]
7	Extrav.	I talk to a lot of different people at parties.	8	Extrav.	I don't like to draw attention to myself.[R]
9	Agreeab.	I am interested in people.[R]	10	Agreeab.	I feel others' emotions.
11	Agreeab.	I have a soft heart.	12	Agreeab.	I am not really interested in others.[R]
13	Agreeab.	I insult people.[R]	14	Agreeab.	I sympathize with others' feelings.
15	Agreeab.	I am not interested in other people's problems.[R]	16	Agreeab.	I feel little concern for others.[R]
17	Agreeab.	I take time out for others.	18	Consc.	I am always prepared.
19	Consc.	I leave my belongings around.[R]	20	Consc.	I pay attention to details.
21	Consc.	I make a mess of things.[R]	22	Consc.	I get chores done right away.
23	Consc.	I often forget to put things back in their proper place.[R]	24	Consc.	I like order.
25	Consc.	I shirk my duties.[R]	26	Consc.	I follow a schedule.
27	Neur.	I am relaxed most of the time.[R]	28	Neur.	I worry about things.
29	Neur.	I am easily disturbed.	30	Neur.	I get stressed out easily.
31	Neur.	I get upset easily.	32	Neur.	I have frequent mood swings.
33	Neur.	I feel threatened easily.[R]	34	Neur.	I often feel blue.
35	Open.	I have a rich vocabulary.	36	Open.	I have difficulty understanding abstract ideas.[R]
37	Open.	I have a good imagination.	38	Open.	I am not interested in abstract ideas.[R]
39	Open.	I have excellent ideas.	40	Open.	I do not have a good imagination.[R]
41	Open.	I am quick to understand things.	42	Open.	I use difficult words.
43	Open.	I spend time reflecting on things.	44	Open.	I am full of ideas.
45	HEXACO-H	I would be tempted to buy stolen goods if I were sure I would not get caught.	46	HEXACO-H	I would be tempted to accept a bribe if I were in a difficult situation.
47	HEXACO-H	I feel entitled to special privileges that others don't have.	48	HEXACO-H	I'd get a lot of pleasure out of being deceitful.
49	HEXACO-H	I'd be tempted to make a lot of money from illegal activities if I could get away with it.	50	Collect.	Personal goals should take precedence over group goals.
51	Collect.	Group harmony and well-being should be prioritized over individual achievement.	52	Collect.	I would do what benefits the group even if it means sacrificing my personal success.
53	Collect.	I should pursue my own goals even if it conflicts with my family's expectations.	54	Intuit.	I prefer to rely on my intuition and gut feelings when making decisions.
55	Intuit.	I prefer to analyze problems systematically and logically before deciding.[R]	56	Intuit.	I like to think through problems carefully before acting.[R]
57	Intuit.	I prefer to take action quickly and adjust as I go.	58	UA	I prefer clear instructions and structured situations over ambiguous and unpredictable ones.
59	UA	I am comfortable taking risks and dealing with uncertainty.[R]	60	UA	I believe in planning for the future rather than living for the present.
61	UA	I believe in living in the moment rather than sacrificing present pleasures for future benefits.[R]			

Under the buggy implementation discussed in Section 4.3 (positive items reversed, negative items left raw), Conscientiousness and Neuroticism appear strongly positively correlated ($r = +0.78$, $n = 43$, with a 95% CI that excludes the human baseline), and the same code reproduces the previously published value of $r = +0.76$ at $n = 33$. Under the corrected reverse-scoring map, the same data yield $r = -0.52$, 95% CI $[-0.71, -0.27]$ at $n = 43$, and $r = -0.42$ at $n = 33$. The corrected value is in the same direction as the human baseline ($r = -0.30$; Soto and John, 2017) and slightly stronger. We treat the corrected $r = -0.52$ as the headline value and report the buggy result alongside it as a methodological note.

Possible explanation: alignment-driven response coupling. Once the C–N artefact is removed, the residual structural deviation visible in the corrected matrix is the over-coupling of A, C, and O, paired with the inversion of Extraversion's relationships with N, C, and O. A plausible source of both is alignment-driven response compression toward a single "positive cluster vs. low-extraversion" axis: items that are socially desirable across dimensions are agreed to together, and items that map onto the loud-extravert pole of the Extraversion subscale are agreed to less often, regardless of their dimension. The HEXACO-H results (Appendix F) and the persona-steering null in Section 4.8 are consistent with this reading.

Beyond the corrected C–N value, Extraversion is negatively linked to Openness ($r = -0.21$ at $n = 43$, in the opposite direction from the human baseline $r = +0.22$), Extraversion-Conscientiousness flips from human $r = +0.11$ to LLM $r = -0.28$, and the C–O correlation is much stronger in LLMs ($+0.72$) than in humans ($+0.10$).

E.2 Convergent Validity

We compare how dimensions relate to each other in LLMs versus published human data (Soto and John, 2017; van der Linden et al., 2010) for ten key pairs under the corrected scoring (Table 15, $n = 43$ models). Six of ten pairs agree in sign, with three of them substantially stronger in LLMs

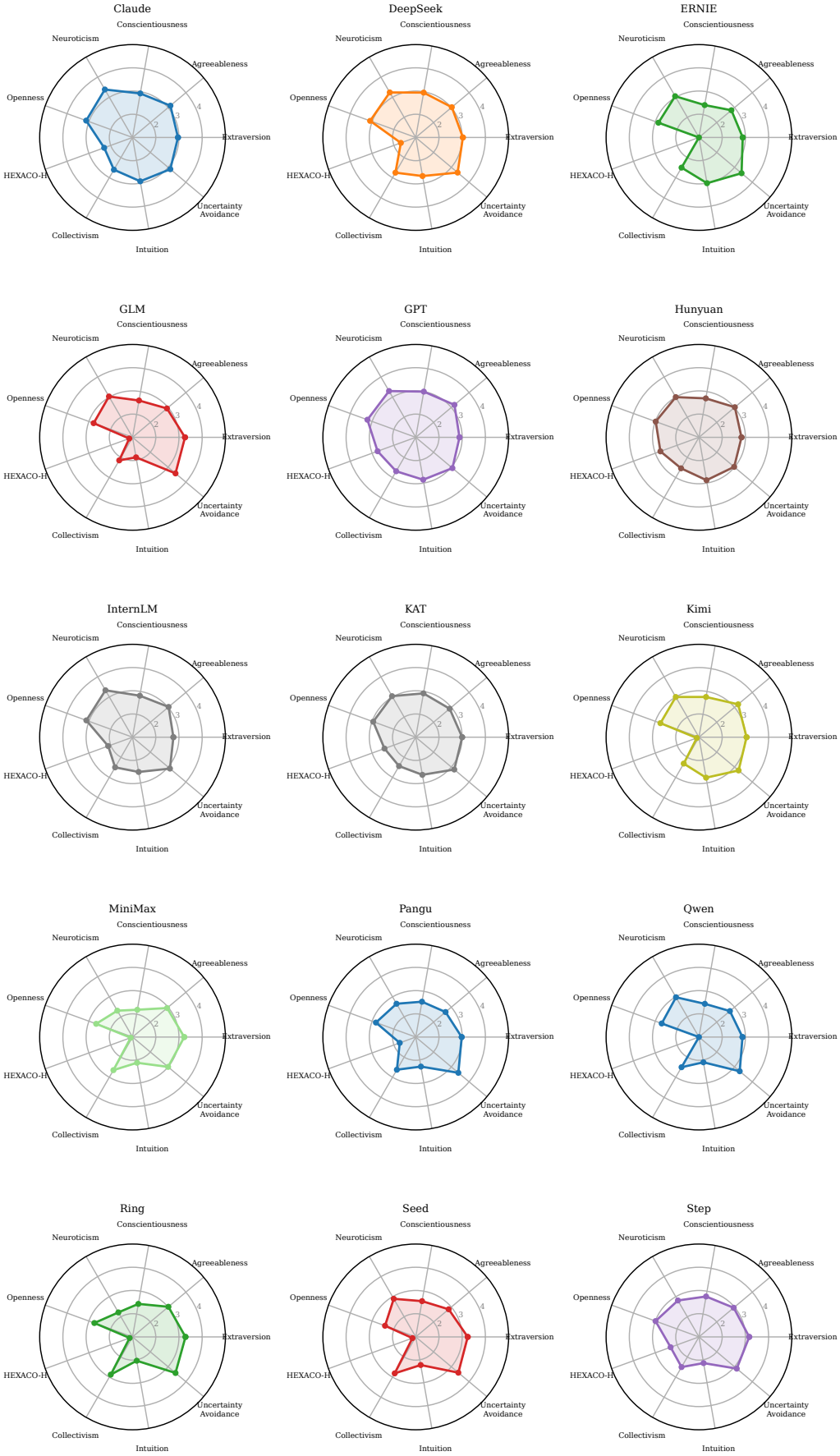

Figure 9: Individual model family radar profiles across 9 dimensions.

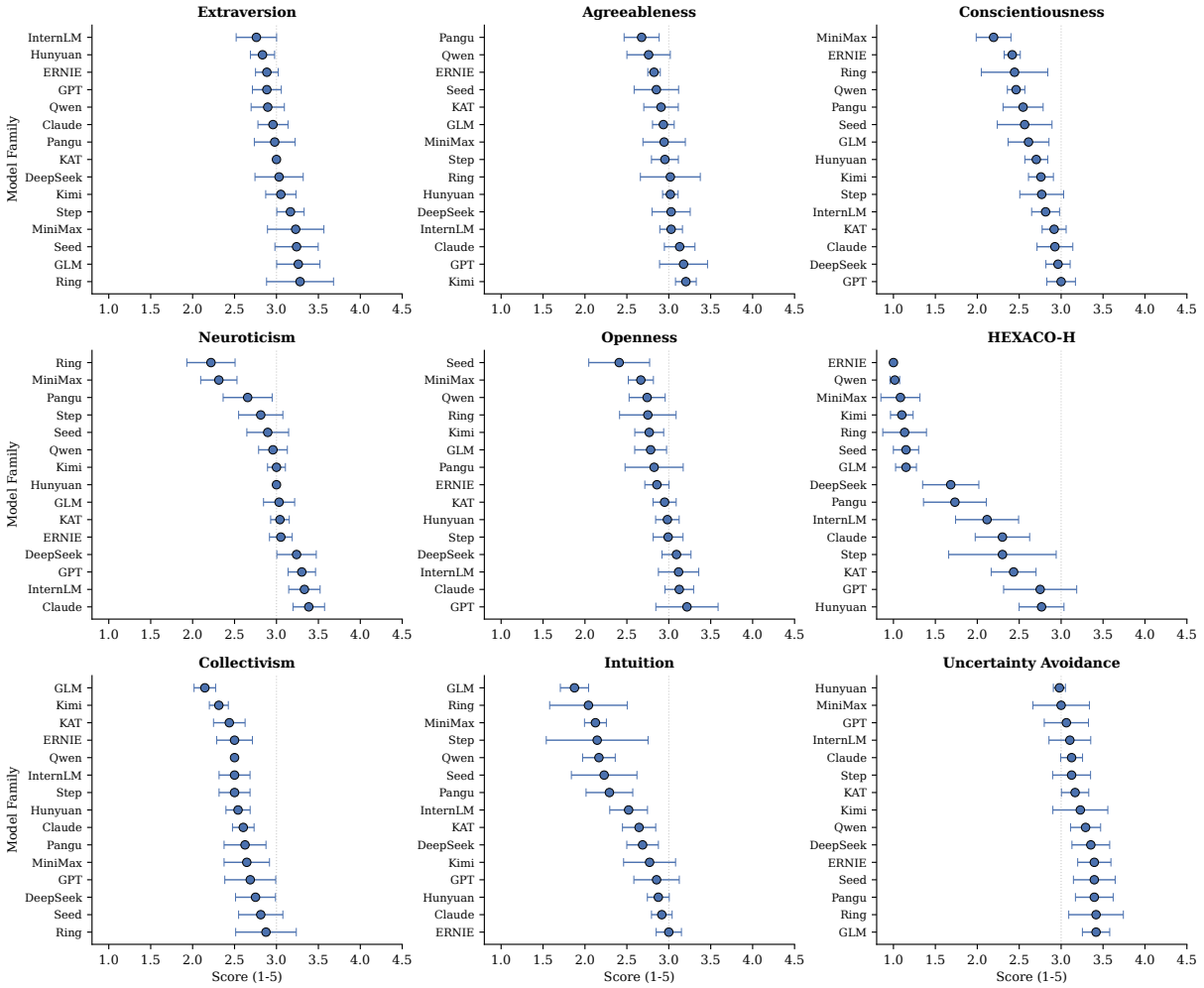

Figure 10: Per-dimension model family mean scores (SD) across all 15 model families. Each panel shows one dimension with families sorted by mean score. Error bars: σ across 12 seeds.

(the A-C, A-O, C-O over-coupling described in Section 4.3). Three pairs flip sign relative to the human baseline, and all three involve Extraversion (E-N, E-C, E-O). The fourth deviation is the A-N pair, which retains the same sign as the human baseline but is much weaker (LLM $r = +0.07$ vs. human $r = -0.25$). Under the buggy reverse-scoring of Schaeckermann et al. (2024) and similar implementations, three additional pairs flip sign (notably C-N), and the headline conclusion of an LLM "personality reversal" is overstated. We re-emphasise that the table reports corrected values; the buggy values are listed in Appendix L for direct comparison.

F HEXACO-H: Supplementary Results

Note: HEXACO-H has unacceptably low internal consistency ($\alpha = 0.27$, 5 items). Results in this section are reported for completeness only and are excluded from all primary analyses.

Table 15: Convergent validity: LLM versus human inter-dimension correlations (Soto and John, 2017), after correcting the BFI reverse-scoring loop. $n = 43$ models from Studies 1, 2, and 6.

Dimension Pair	Human r	LLM r	Status
E vs A	$+0.18$	$+0.18$	matched
E vs C	$+0.11$	-0.28	sign flip
E vs N	-0.34	$+0.53$	sign flip
E vs O	$+0.22$	-0.21	sign flip
A vs C	$+0.29$	$+0.58$	same, stronger
A vs N	-0.25	$+0.07$	weakened
A vs O	$+0.19$	$+0.71$	same, stronger
C vs N	-0.30	-0.52	same, stronger
C vs O	$+0.10$	$+0.72$	same, stronger
N vs O	-0.08	-0.29	same, stronger

Dense models average $H = 1.91$ vs. MoE models $H = 1.31$ ($d = 0.91$, 95% CI $[0.66, 1.20]$, $p < 0.001$; Table 5), with 10 Dense and 5 MoE families. However, architecture and provider effects cannot be disentangled.

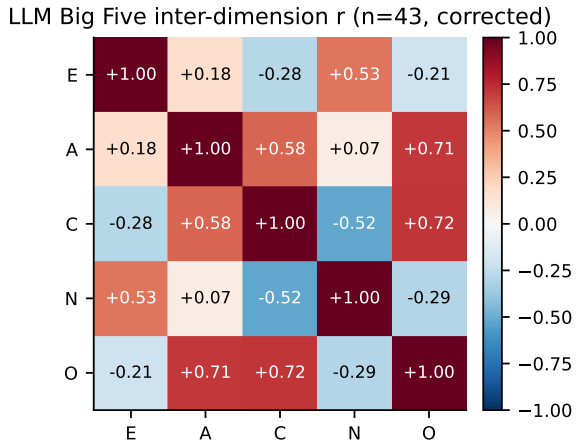

Figure 11: Inter-dimension correlation matrix of Big Five means across 43 models (Study 1 plus Study 2 plus Study 6), under corrected reverse-scoring. E = Extraversion, A = Agreeableness, C = Conscientiousness, N = Neuroticism, O = Openness. The C-N cell is -0.52, same direction as the human BFI-2 baseline of -0.30.

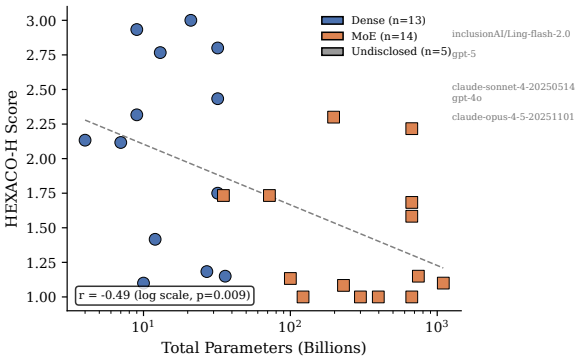

Figure 12: HEXACO-H vs. total parameters across all tested models (27 with disclosed sizes + 4 undisclosed). Colors indicate architecture type. $\alpha = 0.27$; shown for completeness only.

In the broader 27-model sample, H correlates negatively with log-transformed parameters ($r = -0.49$, $p = 0.009$; Figure 12). HEXACO-H is also included in the scale–dimension correlation table (Table 16) and regression table (Table 17) for completeness, but no inferential claims are based on these results.

G Scale–Dimension Correlations

Table 16 reports Pearson correlations between total parameters and each dimension across 27 models with disclosed sizes (including Study 2 within-family variants). Figure 13 shows the corresponding scatter plots for the 15 representative models ($n = 13$ with disclosed parameters).

Table 16: Pearson correlation between total parameters and each dimension (27 models with disclosed sizes, including Study 2 within-family variants). $^{*}p < 0.05$, $^{**}p < 0.01$.

Dimension	r (linear)	r (log scale)
Extraversion	+0.17	+0.24
Agreeableness	+0.42*	+0.15
Conscientiousness	−0.16	−0.42*
Neuroticism	−0.10	−0.38
Openness	−0.25	−0.35
HEXACO-H	−0.38	−0.49**
Collectivism	−0.03	+0.03
Intuition	−0.21	−0.43*
UA	+0.35	+0.42*

H Multivariate Regression: Scale and Architecture

To jointly assess whether model scale and architecture predict response style variation, we fit OLS regressions of the form:

$$Y_d = \beta_0 + \beta_1 \log_{10}(\text{params}) + \beta_2 \cdot \text{arch}_{\text{MoE}} + \epsilon \quad (2)$$

for each dimension d, across the 13 Study 1 models with disclosed parameter counts (GPT-5 and Claude Opus 4.5 excluded).

Table 17: OLS regression: dimension score $\sim \log_{10}(\text{params}) +$ architecture ($n = 13$). Dense = 0, MoE = 1. No effect survives FDR correction across 18 tests (9 dimensions $\times$ 2 predictors).

Dimension	β_{log}	p	β_{arch}	p	R^2
Extraversion	+0.088	0.539	+0.022	0.914	0.162
Agreeableness	+0.121	0.296	−0.174	0.297	0.116
Conscientiousness	+0.183	0.295	−0.399	0.127	0.238
Neuroticism	+0.397	0.109	−0.745	0.045	0.349
Openness	−0.110	0.509	+0.099	0.676	0.052
HEXACO-H	−0.450	0.302	−0.212	0.730	0.414
Collectivism	−0.223	0.181	+0.241	0.307	0.177
Intuition	+0.181	0.543	−0.446	0.307	0.126
UA	+0.060	0.648	+0.056	0.771	0.165

HEXACO-H shows the highest explained variance ($R^2 = 0.41$), driven primarily by architecture (architecture-only $R^2 = 0.34$). Neuroticism shows the largest improvement from adding log-parameters ($\Delta R^2 = +0.20$). However, with only 13 observations and 2 predictors, the regression is severely underpowered; no effect survives FDR correction, and these results should be treated as exploratory.

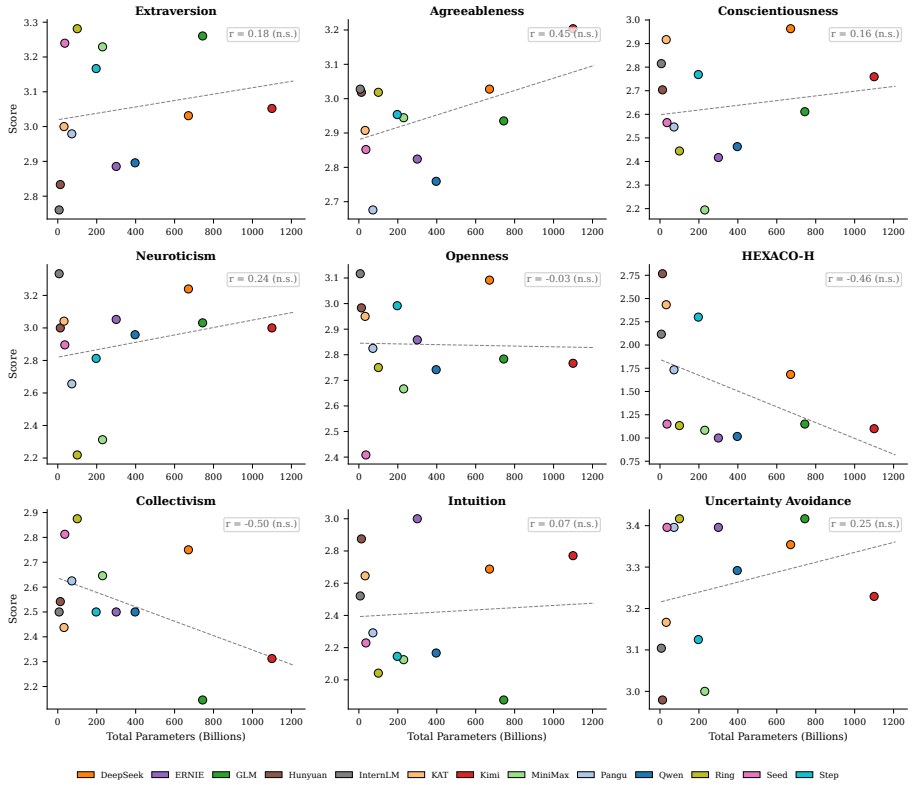

Figure 13: Total parameters vs. dimension score for all 9 dimensions ($n = 13$ models with disclosed total parameters). Red labels: $p < 0.05$; gray labels: not significant.

I Thinking Mode Ablation

As partial evidence for alignment-driven compression, we compare standard chat mode with extended reasoning ("thinking") mode for 4 families (Qwen, DeepSeek, GLM, Kimi). Thinking mode asks the model to reason step-by-step before answering, which engages chain-of-thought and may change how the model handles self-description questions. Each model was tested with 3 seeds in thinking mode, compared against 12-seed chat baselines from Study 1.

Two consistent patterns emerge across all four models. Conscientiousness drops in thinking mode (d from -2.11 to -8.02), suggesting that reasoning pushes the "diligent" response further toward the socially desirable endpoint. Neuroticism also drops (d from -3.74 to -7.41), as if models reason themselves toward emotional stability. HEXACO-H hits the theoretical floor (1.00) in thinking mode for all models, including DeepSeek where chat-mode H is 1.68. Intuition jumps sharply in thinking mode (up to $d = +13.09$ for GLM), suggesting reasoning models claim greater intuitive ability.

These patterns fit alignment-driven compression: when models reason step-by-step, they give more extreme answers on morally and socially loaded dimensions, pushing further toward the endpoints that alignment training targets. However, this comparison mixes inference mode with response length and reasoning effort, and should not be equated with a direct base-vs.-aligned comparison.

Table 18: Thinking mode ablation: mean (SD) for chat vs. thinking mode. Cohen's d computed with pooled SD.

		Chat (Study 1)		Thinking			
Model	Dim.	Mean	SD	Mean	SD	Δ	d
Qwen	C	2.46	0.10	1.70	0.05	−0.76	−8.02
	N	2.96	0.17	2.25	0.00	−0.71	−4.54
	H	1.02	0.06	1.00	0.00	−0.02	−0.36
	Int	2.17	0.19	3.50	0.00	+1.33	+7.61
DeepSeek	C	2.96	0.14	2.30	0.46	−0.66	−2.11
	N	3.24	0.24	2.54	0.06	−0.70	−3.74
	H	1.68	0.34	1.00	0.00	−0.68	−2.63
	O	3.09	0.17	2.27	0.12	−0.82	−5.40
GLM	C	2.61	0.24	1.33	0.16	−1.28	−5.87
	N	3.03	0.19	2.38	0.10	−0.65	−3.91
	H	1.15	0.12	1.00	0.00	−0.15	−1.51
	Int	1.88	0.17	3.92	0.12	+2.04	+13.09
Kimi	C	2.76	0.15	1.81	0.29	−0.95	−5.27
	N	3.00	0.11	2.25	0.00	−0.75	−7.41
	H	1.10	0.13	1.00	0.00	−0.10	−0.84
	Int	2.77	0.31	3.42	0.12	+0.65	+2.24

J Subgroup Analysis: Family, Scale, and Architecture

To check that midpoint compression is not just an artifact of mixing different models, we analyze the expanded sample ($n = 43$ models from Studies 1, 2, and 6) by family, parameter scale, and architecture.

J.1 Within-Family C–N Correlation

Three families contribute 5+ models, enabling within-family correlation analysis under corrected scoring.

Table 19: Within-family C–N correlation and response profile (expanded sample, corrected scoring). Distance = mean absolute deviation from Likert midpoint (3.0) across the 5 BFI dimensions.

Family	n	C–N r	p	Distance	Family C mean	Family N mean
DeepSeek	5	−0.971	0.006	0.439	3.29	2.95
GLM	6	−0.843	0.035	0.503	3.35	2.72
Qwen	5	−0.535	0.353	0.328	3.40	2.97

The negative C–N link is not an artifact of mixing different families. Within DeepSeek alone, C–N reaches $r = -0.97$ ($p = 0.006$); within GLM, $r = -0.84$ ($p = 0.04$); within Qwen, $r = -0.54$ ($p = 0.35$, $n = 5$). All three families reproduce the human-direction C–N trade-off, and DeepSeek does so more strongly than the cross-family pooled estimate of $r = -0.52$ in Section 4.3. This suggests that the human-direction trade-off is a property of LLM responses generally, not just of cross-family heterogeneity, and that within-family version differences (base vs. aligned, reasoning vs. chat) accumulate the same negative-correlation pattern as cross-family differences.

J.2 Response Compression by Scale and Source

We examine whether model scale or source type predicts distance from the Likert midpoint.

Two patterns emerge. First, closed-source models (GPT-5, Claude Opus 4.5) are the most compressed (distance = 0.168 vs. 0.272 for open-source), consistent with more intensive alignment producing more uniform answers. Second, MoE models are *less* compressed (distance = 0.334) than Dense models (0.235), mainly due to lower Conscientiousness (2.57 vs. 2.75), a pattern consistent with the HEXACO-H architecture effect in Section 4.2.

Table 20: Response compression by subgroup. Distance = mean $|score - 3.0|$ across 8 dimensions. Lower distance indicates stronger compression toward the midpoint.

		BFI Mean Scores					
Subgroup	n	Dist.	E	A	C	N	O
Closed-source	2	0.168	2.92	3.15	2.96	3.34	3.17
Open-source	31	0.272	3.06	2.94	2.68	3.00	2.94
Small (<10B)	4	0.254	3.02	2.88	2.80	3.17	3.13
Medium (10–40B)	7	0.247	3.04	2.95	2.68	2.98	2.89
Large (>40B)	4	0.238	2.89	2.93	2.57	3.20	3.04
Dense	21	0.235	3.03	2.97	2.75	3.03	2.98
MoE	10	0.334	3.09	2.89	2.57	2.94	2.91

Model size shows no significant linear link to midpoint distance ($r = -0.24$, $p = 0.38$, $n = 15$), though the trend is slightly negative (larger models slightly more compressed), consistent with the closed-source finding. All subgroups stay near the midpoint (mean distance 0.17–0.33 on a 1–5 scale), confirming that midpoint compression is universal rather than an artifact of model mixing.

K Within-Family Response Style Trajectories

Figure 14 shows response style trajectories across model versions for three families.

Qwen contributes 6 versions (4B to 397B), DeepSeek contributes 4 versions (V2.5 to R1), and GLM contributes 6 versions (GLM-4 to GLM-5). Most dimensions show moderate variation across versions without clear monotonic trends. Notable exceptions: DeepSeek shows the steepest HEXACO-H collapse of any family, from V2.5 ($H = 3.00$) to V3.2 ($H = 1.68$), a drop of 1.32 points consistent with intensified safety alignment between versions. GLM shows the largest within-family shift on Intuition (Cognitive Style), declining from 5.0 to 3.875 across versions. These trajectories are descriptive (2–6 data points per family) and are included to motivate future controlled experiments with more data points per trajectory.

L Reverse-Scoring Audit

The Big Five subscales in our 61-item battery contain four reverse-keyed items per dimension for E, C and O, four for A, and one for N. The implementation we initially used encoded the reverse positions as a zero-indexed list per dimension, then iterated:

```
for rev_idx in BFI_REVERSE[trait]:
    if rev_idx - 1 < len(items[key]):
```

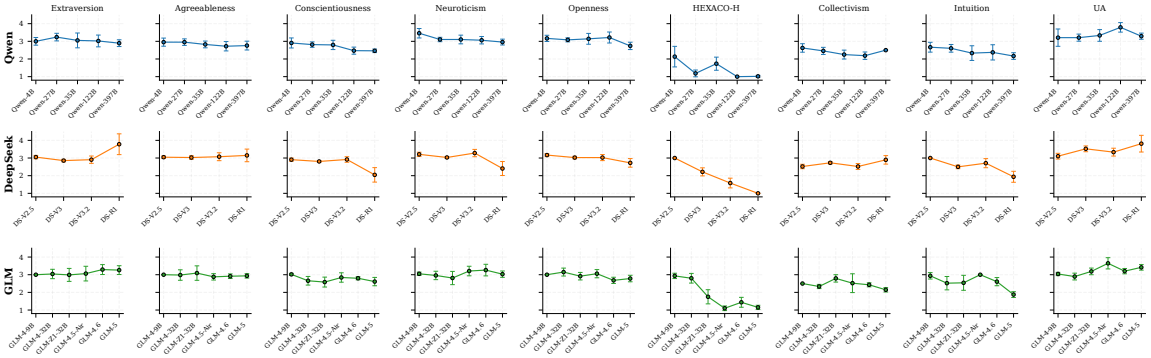

Figure 14: Within-family response style trajectories across model versions. Each row is one model family (Qwen, DeepSeek, GLM); each column is one dimension. Error bars show σ across seeds.

```
items[key][rev_idx - 1] = 6 - items[key][rev_idx - 1]
```

The intended behavior was to flip negative-keyed items at zero-indexed positions $\{1, 3, 5, 7\}$. The actual behaviour, due to the unintended `rev_idx - 1`, was to flip the positive-keyed items at positions $\{0, 2, 4, 6\}$. For Extraversion, Conscientiousness, and Openness this fully inverts the dimension; for Agreeableness it produces a mixed map (some reversals correct, others wrong); for Neuroticism, the reverse-list `[0, 6]` maps to Python negative-index -1 (the last item) plus index 5, none of which correspond to the only true reverse item at position 0.

We undo this artefact in two passes. First, we reconstruct raw 1–5 ratings from the saved values by re-flipping the items the buggy code touched (Table 21). Second, we reapply the corrected reverse-keyed list and recompute dimension means. The reconstruction script (`rescore_existing.py`) is released with the paper. The HEXACO-H subscale, all five items of which are violation-of-ethics statements, is also recomputed in the corrected version as $\text{HEX} = 6 - \overline{x}$ to align with the standard HEXACO direction (high score = high Honesty-Humility); the originally saved value treated raw agreement as the score, which inverts the direction.

Table 21: Buggy versus corrected reverse-scoring positions per Big Five subscale (zero-indexed). Items at the buggy positions were flipped by the original code, leaving the actual negative-keyed items unflipped.

Dimension	List in code	Buggy positions flipped	Correct positions
Extraversion	{1,3,5,7}	{0,2,4,6}	{1,3,5,7}
Agreeableness	{3,4,6,7}	{2,3,5,6}	{3,4,6,7}
Conscientiousness	{1,3,5,7}	{0,2,4,6}	{1,3,5,7}
Neuroticism	{0,6}	$\{-1, 5\}$	{0}
Openness	{1,3,5}	{0,2,4}	{1,3,5}

We also note two model entries in the frontier replication cohort that returned only error responses through the unified LLM gateway during our window: GPT 5.4 and Doubao-Seed 1.6. For these two entries the raw response field stores the literal string `ERROR`, the parser falls back to the midpoint default of 3, and the resulting dimension scores are therefore artefacts of the parser, not of the model. We exclude both entries from the main statistics and report them here for completeness.

M MBTI Persona Prompts

Each persona prompt follows the schema below, with the type-specific fields filled in from McCrae and Costa (1992) and the MBTI Manual (Jiang and Althoff, 2023) (the latter citation is a placeholder; researchers should substitute the canonical Myers-Briggs reference of their choice).

> *System: You are role-playing a human with the {TYPE} ({NICKNAME}) personality type. {TYPE} stands for {EXPANSION}. Cognitive function stack: {STACK}. Behavioural tendencies: {TENDENCIES}. Strengths: {STRENGTHS}. Common challenges: {CHALLENGES}. Answer the following questionnaire as this person would. Stay in character. Respond authentically based on how a {TYPE} individual would genuinely feel, not what is socially expected.*
>
> *User: Statement: "{ITEM_TEXT}". Respond with a single number from 1 to 5: 1 = Strongly Disagree, ..., 5 = Strongly Agree.*

A worked example for INTJ:

> TYPE = INTJ; NICKNAME = The Architect; EXPANSION = Introverted, Intuitive, Thinking, Judging; STACK = Ni-Te-Fi-Se; TENDENCIES = strategic, independent, long-horizon planning, privately intense, reserved; STRENGTHS = systems thinking, decisive execution, high standards; CHALLENGES = dismissive of inefficient processes, emotionally guarded.

The full per-type field table is released with the paper as `docs/mbti_persona_cards.md`.